

Internet Protocols

Learning Objectives

- Define what an internet protocol is
- Explain the role of IP addresses and URLs
- Understand how HTTP and HTTPS work

Engage (5-7 minutes)

When you send a letter, it needs an address, a pin code, and a postal system to deliver it. The internet works the same way—every device has an address (IP), every website has a name (URL), and protocols like HTTP are the postal rules that deliver web pages to your screen.

Explore (10-12 minutes)

Type a website address like `www.google.com` in a browser. Look at the address bar—what do you see? Write down the full URL. Now ask your teacher to show you an IP address using a command. Compare the URL and IP address.

Explain (10-12 minutes)

An IP address is a unique number assigned to every device on a network, like a house number on a street. A URL is the human-readable name of a website, like `www.example.com`. HTTP (HyperText Transfer Protocol) sends web pages without encryption, while HTTPS adds security by encrypting data—like sending a letter in a sealed envelope versus a postcard.

Elaborate (8-10 minutes)

Think of the panchayat office. Each office has a unique address for mail delivery. Similarly, each computer on the internet has an IP address. The URL is like the office name—“Gram Panchayat Office”—while the IP is like its GPS coordinates. Why do you think HTTPS is important when banking online?

Evaluate (5-8 minutes)

Exit ticket: Write the difference between an IP address and a URL. Then explain why HTTPS is safer than HTTP. If you finish early, look up the IP address of any website using an online tool.